Progress Check

1. When two polynomials of same degree has to be divided, _____ should be considered to fix the dividend and divisor.
2. If $r(x) = 0$ when $f(x)$ is divided by $g(x)$ then $g(x)$ is called _____ of the polynomials.
3. If $f(x) = g(x)q(x) + r(x)$, _____ must be added to $f(x)$ to make $f(x)$ completely divisible by $g(x)$.
4. If $f(x) = g(x)q(x) + r(x)$, _____ must be subtracted to $f(x)$ to make $f(x)$ completely divisible by $g(x)$.

Example 3.10 Find the GCD of the polynomials $x^3 + x^2 - x + 2$ and $2x^3 - 5x^2 + 5x - 3$.

Solution Let $f(x) = 2x^3 - 5x^2 + 5x - 3$ and $g(x) = x^3 + x^2 - x + 2$

$$
\begin{array}{r|l}
 & 2 \\
\hline
x^3 + x^2 - x + 2 & 2x^3 - 5x^2 + 5x - 3 \\
 & 2x^3 + 2x^2 - 2x + 4 \\
 & (-)\ \ (-)\ \ \ (+)\ \ (-) \\
\hline
 & -7x^2 + 7x - 7
\end{array}
$$

$$= -7(x^2 - x + 1)$$

$-7(x^2 - x + 1) \neq 0$, note that -7 is not a divisor of $g(x)$

Now, dividing $g(x) = x^3 + x^2 - x + 2$ by the new remainder x^2–x+1 (leaving the constant factor), we get

$$
\begin{array}{r|l}
 & x + 2 \\
\hline
x^2 - x + 1 & x^3 + x^2 - x + 2 \\
 & x^3 - x^2 + x \\
 & (-)\ \ (+)\ \ (-) \\
\hline
 & 2x^2 - 2x + 2 \\
 & 2x^2 - 2x + 2 \\
 & (-)\ \ (+)\ \ (-) \\
\hline
 & 0
\end{array}
$$

Here, we get zero remainder.

Therefore, GCD$(2x^3 - 5x^2 + 5x - 3,\ x^3 + x^2 - x + 2) = x^2 - x + 1$.

Example 3.11 Find the GCD of $6x^3 - 30x^2 + 60x - 48$ and $3x^3 - 12x^2 + 21x - 18$.

Solution Let, $f(x) = 6x^3 - 30x^2 + 60x - 48 = 6(x^3 - 5x^2 + 10x - 8)$ and

$g(x) = 3x^3 - 12x^2 + 21x - 18 = 3(x^3 - 4x^2 + 7x - 6)$

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